Infosys — Aptitude Test 2

Questions 26–50 of 100 · 25 questions

Question 26. Tanya's grandfather was 8 times older to her 16 years ago. He would be three times of her age 8 years from now. Eight years ago what was ratio of Tanya’s age to her grandfather?

Solution:
Let present age of Grandfather and
Tanya be G and T respectively.
G – 16 = 8(T – 16)
G = 8T – 112
---(1)
G + 8 = 3(T + 8)
G = 3T + 16
---(2)
From (1) and (2)
8T – 112 = 3T + 16
T = 128/5 and G = 464/5
8 years ago, T – 8 = 88/5 and
G – 8 = 424/5
Ratio is 88 : 424 = 11 : 53

Question 27. A person is 80 years old in 490 and only 70 years old in 500 in which year is he born?

Solution:
This is only possible when the person
was born in B.C (Before Christ).
The years will proceed in descending
order.
The person was born on 490 + 80
= 570 B.C

Solution:
15 Men
= 10 Women.
1 Man
= (10/15) Woman
5 men = 5(10/15) = (10/3) Women
5 Men and 4 Women
= (10/3) + 4 Women
= 22/3 Women
Time
and
resource
are
inversely
proportional
(10 W)/[(22/3) W) = x days/55 days
x = 75 days

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Question 28. If A can copy 50 pages in 10 hours and A and B together can copy 70 pages in 10 hours, how much time does B takes to copy 26 pages?

Question 29. Sixty men complete a work in 25 days. One man starts working it at and thereafter one more man joins him every day. In how many days the work will be completed?

Solution:
One man can complete the work in
25 x 60 = 1500 days.
Work done in day 1 = 1/1500
Work done in day 2 = 2/1500
Work done in day 3 = 3/1500
Let us assume that the work has been
done for n days.
(1/1500) + (2/1500) + (3/1500) + …
(n/1500) = 1
Hence, 1 + 2 + 3 + … + n = 1500
n(n+1)/2
=1500(sum
of
n
natural
numbers)
n(n+1) = 3000
n ■54
It takes 54 days to complete the work.

Question 30. 15 men or 10 women complete work in 55 days. Then 5 men and 4 women complete work in how many days?

Question 31. If 5/2 artists make 5/2 paintings using 5/2 canvases in 5/2 days then how many artists are required to make 25 paintings using 25 canvases in 25 days?

Solution:
Work done by A in 1 hour = 5 pages
Work done by A and B together in 1 hour
= 7 pages.
Work done by B in 1 hour = 7 – 5
= 2 pages.
Time taken for B to copy 26 pages
= 26/2
= 13 hours.

Solution:
This is a tricky puzzle.
The same number of artists are required
to paint 25 paintings in 25 days using 25

canvases.
To understand,
The work done by 5/2 artists in 1 day is
1 painting.
This is the same case for 25 paintings in
25 days.
Answer is 5/2 artists.

Question 32. In a grass field if 40 cows could eat for 40 days and 30 cows for 60 days, how long could 20 cows eat?

Solution:
Resource
and
time
are
inversely
proportional
Resources = number of cows
Time = number of days
[R1/R2] = [T2/T1]
40/20 = T2/40
T2 = 80 days.

Question 33. At 20% discount, a cycle is sold at a selling price of 2500 Rs. What is the marked price?

Solution:
Selling price =[(100-20)/100]x Marked
price
2500 = (80/100)Marked price
Marked Price = 2500x100/80=Rs. 3125

Question 34. In a certain office, 1/3 of the workers are women, 1/2 of the women are married and 1/3 of the married women have children. If 3/4th of the men are married and 2/3rd of the married men have children, what part of workers are without children?

Solution:
No.of employees = x
No.of women = x/3; No.of men = 2x/3
No.of married women=(x/3)(1/2)=x/6
No.of women with child
= (x/6)(1/3)=x/18
No.of married men = (2x/3)(3/4) = x/2
No.of men with child
= (x/2)(2/3) = x/3
No.of employee with children
=(x/18)+(x/3)
= 7x/18

Part of workers without child
= 1-(7/18)
= 11/18

Question 35. Fifty percent of the articles in a certain magazine are written by staff members. Sixty percent of the articles are on current affairs. If 75 percent of the articles on current affairs are written by staff members with more than 5 years experience of journalism, how many of the articles on current affairs are written by journalists with more than 5 years experience? 20 articles are written by staff members. Of the articles on topics other than current affairs, 50 percent are by staff members with less than 5 years experience.

Solution:
Total number of articles = 20
Number of articles on current affairs
= 60% of 20
= (60/100)20
= 12
Number of articles in current affairs by more than 5 years experienced journalists
= 75% of 12
= 9

Question 36. Two numbers are respectively 20% and 50% more than a third number. The ratio of the two numbers is:

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Solution:
Assume the third number = 100
First number = 120% of 100 = 120
Second number = 150% of 100 = 150
Ratio between first and second number
= 120 : 150
= 4 : 5

Question 37. In an exam 49% candidates failed in English and 36% failed in Hindi and 15% failed in both subjects. If the total number of candidates who passed in English alone is 630. What is the total

number of candidates appeared in exam?

Solution:
Percentage of students failed in English alone
= 49 – 15 = 34
Percentage of students failed in Telugu
= 36 – 15 = 21%
21% students who are fail in only Telugu are the ones passed only in English.
(21/100) total students = 630;
Total = 3000

Question 38. Suresh invested a sum of Rs. 15000 at 9 percent per annum Simple interest and Rs. 12000 at 8 percent per annum compound interest for a period of 2 years. What amount of interest did Suresh earn in 2 years?

Solution:
SI = PNR/100
SI = 15000x2x9/100 = 2700
CI = P[1+(R/100)]2 – P
CI=12000(108/100)(108/100) – 12000
CI = 1996.8
Total interest = 4696.8

Question 39. A town have a population of 500000 and 42% of males and 28% of females are married to same town. Find the total number of males.

Solution:
Since the males and females get married within their town, the number of married males is equal to number of married females.
Therefore,
42% of males
= 28% females
Males : Females
= 28 : 42
= 2 : 3
Number of males = (2/5)500000

Question 40. Present population of town is 35,000 having males and females. If The population of males is increased by 6% and if the population of females is increased by 4%, then after 1 year the population becomes 36,700. Find the number males and females ?

Solution:
Let number of Males be X
Number of Females = 35000 – X
At the end of one year,
Number of Males = 1.06X
Number of Females = 1.04(35000-X)
1.06X + 36400 – 1.04X = 36700
0.02X = 300
X = 15000 = Number of males
35000–15000 = 20000 = Number of females

Question 41. A shopkeeper purchased an article at 20% discount on list price, he marked up his article in such a way that after selling the article at 20% discount, he gained 20% on SP. what % is SP of the list price?

Un-grazed area = Area(Square)– Area(circle)
= 632 – πx632/4
= 632(1 – (π/4)) = 3969(1-(3.14/4))
= 853 sq.m

Solution:
Assume list price = 100
Shop keeper purchased at Rs. 80
To get 20% profit, selling price
= 1.2x80 = 96
To get Rs. 96 after 20% discount,
0.8M.P = 96;
Marked Price = 120.
% of S.P with respect to list price
= (96/100)100
= 96%

Question 42. An electric wire runs for 1 km between some number of poles. If one pole is removed the distance between each pole increases by 1 2/6 (mixed fraction). How many poles were there initially?

I n f o s y s

Solution:
Assume that the number poles initially was ‘n’.
Number of gaps when there is n poles = n-1
Distance of each gap = 1/(n-1)
The number of poles after removing 1 pole = n-1
Number of gaps = n-2
Distance between each gap = 1/(n-2)
Given that
1/(n-2) = 8/6[1/(n-1)]
By solving, we get
6n – 6 = 8n – 16
10 = 2n
n = 5, there were 5 poles initially.

= 200000

Question 43. 4 horses are tethered at 4 corners of a square plot of side 63 meters so that they just cannot reach one another. The area left un-grazed is?

Solution:
The shaded area in the below diagram represents the grazed area.

Question 44. A rabbit is tied to one end of an equilateral triangle of side 5 m with a rope length of 8m. The rabbit is not allowed to travel inside the triangle then find the maximum area covered by the rabbit?

63

Solution:
The below diagram explains the question.

To find
Area(circle)-Area(triangle)
= πx82 – [(√3/4)x52]
= 64π – 10.8

= 190.2 sq.m

Question 45. Two parallel chords of length 32 and 24 cm .and radius of circle is 20cm. Find the distance between the chords.

Solution:
Observe the diagram below.

32

16

a
b
c

20

o

20

f

12

e
d

24

O is the center of the circle.
‘ab’ and ‘cd’ are the chords with length
32 and 24 respectively.
‘oac’ and ‘odf’ are right angled triangles.
(of)2 = 202 – 122 ■‘of’ = 16
(oc)2 = 202 – 162 ■‘oc’ = 12
Distance between two chords = of + oc
= 16 + 12 = 28

Question 46. when a circle is inscribed in a square ,and that square is inscribed in a circle. Then ratio of big circle to small circle is:

Solution:
Observe the diagram.

I n f o s y s

di

do

If side of the square is A,
Then diameter of inner circle = A
Diameter of outer circle = diagonal of the square
Diagonal of square = A√2
Diameter of outer circle = A√2
Ratio between two circles
= A√2 : A
= √2 : 1

Question 47. Mr. Lloyd wants to fence his Square shaped land of 120m each side. If a pole is to be laid every 12m how many poles will he need?

Solution:
Number of corner poles = 4
For 120m length, there are 10 segments of 12m. For 10 segments the number of poles required is 11.
Excluding he corner poles, each side needs 9 poles.
Total number of poles = 4 x 9 + 4 = 40

Question 48. A person walking takes 26 steps to come down on a escalator and it takes 30 seconds for him for walking. The same person while running takes 18 second and 34 steps. How many steps are there in the escalator?

Solution:
Assume that the number of steps in the escalator = N
While walking, the person covers 26

steps, and the escalator covers (N-26) steps in 30 seconds.
While running, the person covers 34 steps and the escalator covers (N-34) steps in 18 seconds.
In both cases, speed of escalator is same.
Speed of escalator = Distance (in steps)/time
Escalator speed in case 1:
S = (N-26)/30
Escalator speed in second case:
S = (N-34)/18
Therefore,
(N-26)/30 = (N-34)/18
18N – 468 = 30N – 1020
12N = 552
Number of steps = N = 46

Solution:
Assume speed of 5 horses = S
Speed of 3 horses = 3S/5
Time taken for 5 horses to cover 50 miles
= 50/S
Time taken for 3 horses to cover the same 50 miles
= [50/(3S/5)]

50/S = [50/(3S/5)] – 24
S = 25/18 miles per hour
After first 24 hours, actual time taken by
the person to reach the destination be ‘t’,
Distance = Speed x time
S x t= (3S/5) x (t + 48)
t = 72 hours
Total time = 72 + 24
= 96 hours
Distance = 400/3 miles

I n f o s y s

Question 49. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. What is the ratio of the speed of the train to that of the car?

Question 50. A person has to cover the fixed distance through his horses. There are five horses in the cart. They ran at the full potential for the 24 hours continuously at constant speed and then two of the horses ran away to some other direction. So he reached the destination 48 hours behind the schedule. If the five horses would have run 50 miles more, then the person would have been only 24 hours late. Find the distance of the destination.